Supriyo Bhattacharya

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B.Tech Electrical Engineering
Indian Institute Of Technology, Jammu

Github | Website
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year of Passing
B.Tech, Electrical Engineering	Indian Institute of Technology, Jammu	7.38 (Current)	July 2026 (expected)
12th Standard	Kendriya Vidyalaya AFS Chabua	94%	2021
10th Standard	Kendriya Vidyalaya AFS Chabua	88.4%	2019

Scholastic Achievements

- Winner of Panasonic Asia Scholarship Program (2022).
- Secured **AIR 8934** in **JEE Advanced** (2022).
- **Chanakya Research Fellowship** on agentic Tsetlin Machine for skin-cancer detection (2024).

Publications

- **Elephant in the Room: Composite Vanilla Tsetlin Machine with Hybrid Feature Extraction**, ACDSA Conference, IEEE Xplore (2026).

Work Experience

- **Panasonic India Innovation Centre** *May 2025 – Aug 2025*
  Data Science Intern Bangalore
  – Engineered big data pipelines using **Apache Spark** for large-scale AC fault prediction.
  – Leveraged sensory IoT data to develop predictive maintenance workflows.
  – Designed a **hybrid ML model** achieving **97% accuracy** and **99% precision**, reducing downtime significantly.

- **Ernst and Young** *May 2024 – Jul. 2024*
  Data Analyst Intern New Delhi
  – Used Python and SQL to perform complex data analysis and generate actionable insights.
  – Analyzed India's trade data from MeiTY to identify key trends and patterns.
  – Worked with senior analysts on diverse projects, applying EY methodologies to develop data-driven strategies.

- **Craving Alpha LLP** *May 2024 – Jun. 2024*
  Research Analyst Intern Kolkata
  – Assisted in developing and testing methods for equity selection and risk management.
  – Implemented tools for data processing, backtesting, and performance analysis using Python.

Projects

- **Fractal Geometry Integration in Tsetlin Machine for Pattern Recognition** *Apr. 2024 – Present*
  IIT Jammu
  – **Tech**: Python, C, Tsetlin Machine, Arduino Nano BLE 33
  – Enhanced Tsetlin Machine by aggregating class sums from multiple TMs for robust predictions.
  – Applied HOG and color thresholding (3x3, 4x4) for multi-perspective pattern recognition.

- **QuantLab - Machine Learning-Based Algorithmic Trading Engine** *Aug. 2024 – Present*
  Sole Developer & Project Lead Website
  – **Tech**: Python, Django, FYERS API, Azure, TimeSeries Classification
  – Built Django-based platform for algorithmic trading with ML-powered price forecasting.
  – Integrated FYERS API for real-time market data and automated trading.

- **EMG Sensor Integration in Below-Knee Prosthetics** *Jan. 2024 – May. 2024*
  IIT Jammu Github
  – **Tech**: Raspberry Pi, Flex Sensor, EMG Sensors, Stepper Motor
  – Designed EMG-based system to capture myographic signals for prosthetic control.
  – Integrated stepper motor and flex sensors for improved movement detection.

- **WaveText - Messaging App** *Sept. 2024*
  Sole Developer & Project Lead Github
  – **Tech**: REST API, WebSocket, Bootstrap CSS
  – Developed Django + WebSocket messaging app with private/group chats and notifications.

Key Courses Taken

- DSA, Computer Networks, DBMS, Machine Learning, Computer Programming, Linear Algebra

Technical Skills

- **Programming:** Python (Pandas, NumPy, PySpark), C++, Bash Scripting
- **ML/AI:** Scikit-learn, TensorFlow, Deep Learning, Data Engineering (Spark)
- **Systems:** Linux, Git, SQL, Azure
- **Web:** Django, REST, WebSockets
- **Finance:** Equity Research, Algo-Trading

Positions of Responsibility

- **Student Representative,** Career & Development Services, IIT Jammu *Present*
- **Class Representative,** IIT Jammu *Oct 2022 - Mar 2023*

Certifications

- Machine Learning and Deep Learning Course by **Udemy**.
- Advanced Data Analytics Professional Certificate by **Google**.
- Cybersecurity Professional Certificate by **Google**.

Additional Information

- **Hobbies:** *Badminton, Table Tennis, Investment and Finance*
- **Languages:** *Bengali, Hindi, English*
- **Extracurricular Activities:** *NSS volunteer — taught evening classes to village children in Jagti (2024)*